



Lecture 9 – Introduction to Deep Learning

30 MCQs (Dr. Ehab Essa Exam Style)

1. In shallow machine learning, feature extraction is:

- A. Automatic
- B. Manual
- C. Random
- D. Optional

Answer: B

2. Deep learning differs from machine learning mainly in:

- A. Dataset size
- B. Feature learning
- C. Output format
- D. Hardware

Answer: B

3. In deep learning, features are learned:

- A. Manually
- B. Randomly
- C. Directly from data
- D. From rules

Answer: C

4. Deep learning models consist of:

- A. One layer
- B. Linear units only
- C. Multiple layers
- D. No activation

Answer: C

5. Deep learning relies on learning:

- A. Flat features
- B. Feature hierarchies
- C. Labels only
- D. Rules

Answer: B

6. Hierarchical compositionality means:

- A. Single computation
- B. Layered representations
- C. Manual features
- D. Linear models

Answer: B

7. AlexNet is associated mainly with:

- A. 2006
- B. 2012
- C. 2018
- D. 2022

Answer: B

8. AlexNet demonstrated the power of:

- A. RNNs
- B. GANs

- C. CNNs
- D. Transformers

Answer: C

9. The AI winter occurred due to limitations of:

- A. CNNs
- B. Transformers
- C. Single-layer perceptrons
- D. GANs

Answer: C

10. Backpropagation was popularized in:

- A. 1950s
- B. 1960s
- C. 1980s
- D. 2000s

Answer: C

11. The perceptron was introduced by:

- A. Hinton
- B. Rosenblatt
- C. Vapnik
- D. Goodfellow

Answer: B

12. ADALINE was introduced by:

- A. McCulloch & Pitts
- B. Widrow & Hoff
- C. Vaswani et al.
- D. He et al.

Answer: B

13. ImageNet challenge focuses mainly on:

- A. Regression
- B. Clustering
- C. Classification
- D. Ranking

Answer: C

14. ImageNet contains approximately:

- A. 10 categories
- B. 100 categories
- C. 1000 categories
- D. 10000 categories

Answer: C

15. An artificial neural network mimics the:

- A. CPU
- B. GPU
- C. Human brain
- D. Memory

Answer: C

16. The connection strength between neurons is called:

- A. Bias
- B. Threshold
- C. Weight
- D. Feature

Answer: C

17. The neuron output is determined using:

- A. Sorting
- B. Activation function
- C. Compilation
- D. Sampling

Answer: B

18. The purpose of activation functions is to introduce:

- A. Linearity
- B. Noise
- C. Non-linearity
- D. Sparsity

Answer: C

19. Bias in a neuron allows:

- A. Input scaling
- B. Output shifting
- C. Feature removal
- D. Weight sharing

Answer: B

20. A neural network is characterized by all EXCEPT:

- A. Architecture
- B. Activation function
- C. Training algorithm
- D. Dataset size

Answer: D

21. Sigmoid activation outputs values between:

- A. -1 and 1
- B. 0 and 1
- C. 0 and ∞
- D. $-\infty$ and ∞

Answer: B

22. Vanishing gradient is a disadvantage of:

- A. ReLU
- B. Softmax
- C. Sigmoid
- D. Max pooling

Answer: C

23. ReLU is bounded:

- A. Above
- B. Below
- C. Both
- D. Neither

Answer: B

24. The “dying ReLU” problem occurs when:

- A. Output > 1
- B. Inputs are negative
- C. Weights increase
- D. Bias is zero

Answer: B

25. Softmax is mainly used for:

- A. Binary classification
- B. Regression
- C. Multi-class classification
- D. Clustering

Answer: C

26. A single neuron can solve:

- A. Any problem
- B. Nonlinear problems
- C. Linearly separable problems
- D. Multi-class tasks

Answer: C

27. Multi-layer neural networks include:

- A. Only input layer
- B. Only output layer
- C. Hidden layers
- D. No weights

Answer: C

28. CNNs are widely used in:

- A. Sorting
- B. Image recognition
- C. Database systems
- D. Compilation

Answer: B

29. Pooling layers are used mainly to:

- A. Increase parameters
- B. Reduce dimensionality

- C. Add non-linearity
- D. Generate labels

Answer: B

30. GANs consist of:

- A. Encoder and decoder
- B. Generator and discriminator
- C. Input and output
- D. CNN and RNN

Answer: B